

Axonal fiber terminations concentrate on gyri.

PMID: 22190432 | Indexed in PubMed

Convolved cortical folding and neuronal wiring are 2 prominent attributes of the mammalian brain. However, the macroscale intrinsic relationship between these 2 general cross-species attributes, as well as the underlying principles that sculpt the architecture of the cerebral cortex, remains unclear. Here, we show that the axonal fibers connected to gyri are significantly denser than those connected to sulci. In human, chimpanzee, and macaque brains, a dominant fraction of axonal fibers were found to be connected to the gyri. This finding has been replicated in a range of mammalian brains via diffusion tensor imaging and high-angular resolution diffusion imaging. These results may have shed some lights on fundamental mechanisms for development and organization of the cerebral cortex, suggesting that axonal pushing is a mechanism of cortical folding.

Neuroanatomic and magnetic resonance imaging references for normal development of cerebral sulci of laboratory primate, cynomolgus monkeys (*Macaca fascicularis*).

PMID: 22348780 | Indexed in PubMed

Cynomolgus monkey (*Macaca fascicularis*) is a popular laboratory primate belonging to Old World monkeys, which are the group most closely related to humans except for the apes. This paper summarizes a series of our studies regarding the development of cerebral sulci and gyri in this primate, and the stated possibility of evaluation of the sulcal development for assessing the developmental toxicity testing. The cerebrum of cynomolgus monkeys experienced a regular sequence of emergence of sulci and gyri on gross observation while such timetables corresponded to those obtained by magnetic resonance imaging (MRI) with a lag time of 10-30 days. When the timetables for the emergence of anatomically identical primary sulci and gyri were compared between cynomolgus monkeys and humans, their chronological sequences were comparable, while some sulci and gyri located on the phylogenetically newer cortical region in humans emerged earlier in monkeys. The present paper further indicates brief procedures for evaluating cerebral abnormalities and/or maturity using brain specimens without MRI measurements. The primary sulcal lengths measured by the 'cotton thread' method were a brief index of the degree of regional gyrification. As the development of a calcarine sulcus was closely correlated with morphological maturation of the lateral ventricle, which changed drastically during embryonic days (EDs) 90-100, the cerebral maturity on ED 100 could be evaluated by the infolding of that sulcus. Thus, the present paper provides gross anatomical and MRI references and brief procedures for investigating the normality of the development of cerebral sulci and gyri of laboratory primates, cynomolgus monkeys.

Anatomy-Guided Spatio-Temporal Graph Convolutional Networks (AG-STGCNs) for Modeling Functional Connectivity Between Gyri and Sulci Across Multiple Task Domains.

PMID: 35930515 | Indexed in PubMed

The cerebral cortex is folded as gyri and sulci, which provide the foundation to unveil anatomo-functional relationship of brain. Previous studies have extensively demonstrated that gyri and sulci exhibit intrinsic functional difference, which is further supported by morphological, genetic, and structural evidences.

Therefore, systematically investigating the gyro-sulcal (G-S) functional difference can help deeply understand the functional mechanism of brain. By integrating functional magnetic resonance imaging (fMRI) with advanced deep learning models, recent studies have unveiled the temporal difference in functional activity between gyri and sulci. However, the potential difference of functional connectivity, which represents functional dependency between gyri and sulci, is much unknown. Moreover, the regularity and variability of the G-S functional connectivity difference across multiple task domains remains to be explored. To address the two concerns, this study developed new anatomy-guided spatio-temporal graph convolutional networks (AG-STGCNs) to investigate the regularity and variability of functional connectivity differences between gyri and sulci across multiple task domains. Based on 830 subjects with seven different task-based and one resting state fMRI (rs-fMRI) datasets from the public Human Connectome Project (HCP), we consistently found that there are significant differences of functional connectivity between gyral and sulcal regions within task domains compared with resting state (RS). Furthermore, there is considerable variability of such functional connectivity and information flow between gyri and sulci across different task domains, which are correlated with individual cognitive behaviors. Our study helps better understand the functional segregation of gyri and sulci within task domains as well as the anatomo-functional-behavioral relationship of the human brain.

Cortical Folding Pattern and its Consistency Induced by Biological Growth.

PMID: 26404042 | Indexed in PubMed

Cortical folding, characterized by convex gyri and concave sulci, has an intrinsic relationship to the brain's functional organization. Understanding the mechanism of the brain's convoluted patterns can provide useful clues into normal and pathological brain function. In this paper, the cortical folding phenomenon is interpreted both analytically and computationally, and, in some cases, the findings are validated with experimental observations. The living human brain is modeled as a soft structure with a growing outer cortex and inner core to investigate its developmental mechanism. Analytical interpretations of differential growth of the brain model provide preliminary insight into critical growth ratios for instability and crease formation of the developing brain. Since the analytical approach cannot predict the evolution of cortical complex convolution after instability, non-linear finite element models are employed to study the crease formation and secondary morphological folds of the developing brain. Results demonstrate that the growth ratio of the cortex to core of the brain, the initial thickness, and material properties of both cortex and core have great impacts on the morphological patterns of the developing brain. Lastly, we discuss why cortical folding is highly correlated and consistent by presenting an intriguing gyri-sulci formation comparison.

Automatic segmentation of human brain sulci.

PMID: 18325826 | Indexed in PubMed

The neocortical surface has a rich and complex structure comprised of folds (gyri) and fissures (sulci). Sulci are important macroscopic landmarks for orientation on the cortex. A precise segmentation and labeling of sulci is helpful in human brain mapping studies relating brain anatomy and function. Due to their structural complexity and inter-subject variability, this is considered as a non-trivial task. An automatic algorithm is proposed to accurately segment neocortical sulci: vertices of a white/gray matter interface mesh are classified under a Bayesian framework as belonging to gyral and sulcal compartments

using information about their geodesic depth and local curvature. Then, vertices are collected into sulcal regions by a watershed-like growing method. Experimental results demonstrate that the method is accurate and robust.

Heterogeneity and transcriptional drivers of triple-negative breast cancer.

PMID: 38100350 | Indexed in PubMed

Triple-negative breast cancer (TNBC) is a heterogeneous disease with limited treatment options. To characterize TNBC heterogeneity, we defined transcriptional, epigenetic, and metabolic subtypes and subtype-driving super-enhancers and transcription factors by combining functional and molecular profiling with computational analyses. Single-cell RNA sequencing revealed relative homogeneity of the major transcriptional subtypes (luminal, basal, and mesenchymal) within samples. We found that mesenchymal TNBCs share features with mesenchymal neuroblastoma and rhabdoid tumors and that the PRRX1 transcription factor is a key driver of these tumors. PRRX1 is sufficient for inducing mesenchymal features in basal but not in luminal TNBC cells via reprogramming super-enhancer landscapes, but it is not required for mesenchymal state maintenance or for cellular viability. Our comprehensive, large-scale, multiplatform, multiomics study of both experimental and clinical TNBC is an important resource for the scientific and clinical research communities and opens venues for future investigation.

Neuroblastoma is composed of two super-enhancer-associated differentiation states.

PMID: 28650485 | Indexed in PubMed

Neuroblastoma and other pediatric tumors show a paucity of gene mutations, which has sparked an interest in their epigenetic regulation. Several tumor types include phenotypically divergent cells, resembling cells from different lineage development stages. It has been proposed that super-enhancer-associated transcription factor (TF) networks underlie lineage identity, but the role of these enhancers in intratumoral heterogeneity is unknown. Here we show that most neuroblastomas include two types of tumor cells with divergent gene expression profiles. Undifferentiated mesenchymal cells and committed adrenergic cells can interconvert and resemble cells from different lineage differentiation stages. ChIP-seq analysis of isogenic pairs of mesenchymal and adrenergic cells identified a distinct super-enhancer landscape and super-enhancer-associated TF network for each cell type. Expression of the mesenchymal TF PRRX1 could reprogram the super-enhancer and mRNA landscapes of adrenergic cells toward a mesenchymal state. Mesenchymal cells were more chemoresistant in vitro and were enriched in post-therapy and relapse tumors. Two super-enhancer-associated TF networks, which probably mediate lineage control in normal development, thus dominate epigenetic control of neuroblastoma and shape intratumoral heterogeneity.

MiDAS 4: A global catalogue of full-length 16S rRNA gene sequences and taxonomy for studies of bacterial communities in wastewater treatment plants.

PMID: 35393411 | Indexed in PubMed

Microbial communities are responsible for biological wastewater treatment, but our knowledge of their diversity and function is still poor. Here, we sequence more than 5 million high-quality, full-length 16S rRNA gene sequences from 740 wastewater treatment plants (WWTPs) across the world and use the sequences to construct the 'MiDAS 4' database. MiDAS 4 is an amplicon sequence variant resolved, full-length 16S rRNA gene reference database with a comprehensive taxonomy from domain to species level for all sequences. We use an independent dataset (269 WWTPs) to show that MiDAS 4, compared to commonly used universal reference databases, provides a better coverage for WWTP bacteria and an improved rate of genus and species level classification. Taking advantage of MiDAS 4, we carry out an amplicon-based, global-scale microbial community profiling of activated sludge plants using two common sets of primers targeting regions of the 16S rRNA gene, revealing how environmental conditions and biogeography shape the activated sludge microbiota. We also identify core and conditionally rare or abundant taxa, encompassing 966 genera and 1530 species that represent approximately 80% and 50% of the accumulated read abundance, respectively. Finally, we show that for well-studied functional guilds, such as nitrifiers or polyphosphate-accumulating organisms, the same genera are prevalent worldwide, with only a few abundant species in each genus.

Genome-wide analyses of 200,453 individuals yield new insights into the causes and consequences of clonal hematopoiesis.

PMID: 35835912 | Indexed in PubMed

Clonal hematopoiesis (CH), the clonal expansion of a blood stem cell and its progeny driven by somatic driver mutations, affects over a third of people, yet remains poorly understood. Here we analyze genetic data from 200,453 UK Biobank participants to map the landscape of inherited predisposition to CH, increasing the number of germline associations with CH in European-ancestry populations from 4 to 14. Genes at new loci implicate DNA damage repair (PARP1, ATM, CHEK2), hematopoietic stem cell migration/homing (CD164) and myeloid oncogenesis (SETBP1). Several associations were CH-subtype-specific including variants at TCL1A and CD164 that had opposite associations with DNMT3A-versus TET2-mutant CH, the two most common CH subtypes, proposing key roles for these two loci in CH development. Mendelian randomization analyses showed that smoking and longer leukocyte telomere length are causal risk factors for CH and that genetic predisposition to CH increases risks of myeloproliferative neoplasia, nonhematological malignancies, atrial fibrillation and blood epigenetic ageing.

Peto's "Paradox" and Six Degrees of Cancer Prevalence.

PMID: 38275822 | Indexed in PubMed

Peto's paradox and the epidemiologic observation of the average six degrees of tumor prevalence are studied and hypothetically solved. A simple consideration, Petho's paradox challenges our intuitive understanding of cancer risk and prevalence. Our simple consideration is that the more a cell divides, the higher the chance of acquiring cancerous mutations, and so the larger or longer-lived organisms have more cells and undergo more cell divisions over their lifetime, expecting to have a higher risk of developing cancer. Paradoxically, it is not supported by the observations. The allometric scaling of species could answer the Peto paradox. Another paradoxical human epidemiology observation in six

average mutations is necessary for cancer prevalence, despite the random expectations of the tumor causes. To solve this challenge, game theory could be applied. The inherited and random DNA mutations in the replication process nonlinearly drive cancer development. The statistical variance concept does not reasonably describe tumor development. Instead, the Darwinian natural selection principle is applied. The mutations in the healthy organism's cellular population can serve the species' evolutionary adaptation by the selective pressure of the circumstances. Still, some cells collect multiple uncorrected mutations, adapt to the extreme stress in the stromal environment, and develop subclinical phases of cancer in the individual. This process needs extensive subsequent DNA replications to heritage and collect additional mutations, which are only marginal alone. Still, together, they are preparing for the first stage of the precancerous condition. In the second stage, when one of the caretaker genes is accidentally mutated, the caused genetic instability prepares the cell to fight for its survival and avoid apoptosis. This can be described as a competitive game. In the third stage, the precancerous cell develops uncontrolled proliferation with the damaged gatekeeper gene and forces the new game strategy with binary cooperation with stromal cells for alimentation. In the fourth stage, the starving conditions cause a game change again, starting a cooperative game, where the malignant cells cooperate and force the cooperation of the stromal host, too. In the fifth stage, the resetting of homeostasis finishes the subclinical stage, and in the fifth stage, the clinical phase starts. The prevention of the development of mutated cells is more complex than averting exposure to mutagens from the environment throughout the organism's lifetime. Mutagenic exposure can increase the otherwise random imperfect DNA reproduction, increasing the likelihood of cancer development, but mutations exist. Toxic exposure is more challenging; it may select the tolerant cells on this particular toxic stress, so these mutations have more facility to avoid apoptosis in otherwise collected random mutational states.
